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Software Verification and Validation (DES3103)

by

Dr. Nor Asiah Mohamad@Razak

Course Learning Outline

1. Briefing Session (Ice Breaking via Whiteboard, Class rules & Pro forma): **M. Teams**
2. Introduction to Software Verification & Validation I & II (V&V)
3. Managing V&V with SQA Integration
4. Quality metrics or attributes in V&V

WEEK 6-MID-TERM TEST (QUIZ): Topic 1 – 3-(CLO1:C2) – 20 MARKS

WEEK 7-PHASE 1: ASSIGNMENT 1-ONLINE PROJECT-BASED LEARNING (OPBL)-(CLO3:A5) - 20 MARKS: M. Teams

WEEK 9-PHASE 1: PRESENTATION & REFLECTION ON PHASE 1: M. Teams

5. Software Reviews – Will be covered on Assignments 1 & 2
6. Quality Control: Software Testing - Will be covered in Assignments 1 & 2
7. Human-Computer Interface Testing - Will be covered in Assignments 1 & 2
8. Manual Testing and Automation Testing: Will be covered in Final Assessment

WEEK 11-PHASE 2: ASSIGNMENT 2-ONLINE PROJECT-BASED LEARNING (OPBL)-(CLO2&4:P4&C5) - 20 MARKS: M. Teams
WEEK 13-PHASE 2: PRESENTATION & REFLECTION ON PHASE 2: M. Teams

9. Quality Maturity Models
10. Software Problem Analysis and Reporting
11. Quality Culture in doing V&V.
12. Safety and Security in V&V

WEEK 14: REFLECTION ON IMPLEMENTATION OPBL (SOFTWARE TESTERS) IN THE V&V COURSE: e-FORUM

WEEK 16-PHASE 3: FINAL ASSESSMENT – INDIVIDUAL SPECIAL PROJECT BASED ON PHASE 2 (CLO4:C5) (2 WEEKS) - 40 MARKS : e-ASSESSMENT

TOPIC 7

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Human-Computer -Interface (HCI) Testing

Human-Computer-Interface (HCI) Testing. The variety of aspects of usefulness and usability, heuristic evaluation, cognitive walkthroughs, user testing approaches (observation method, think-aloud method, etc.)

Testing techniques for web sites and formal experiments to test hypotheses about specific HCI controls.

Three broad categories of computer user:

1) Expert users - detailed knowledge of that particular system.

2) Occasional users - know well how to perform the tasks they need to perform frequently.

3) Novices - never used the system before.

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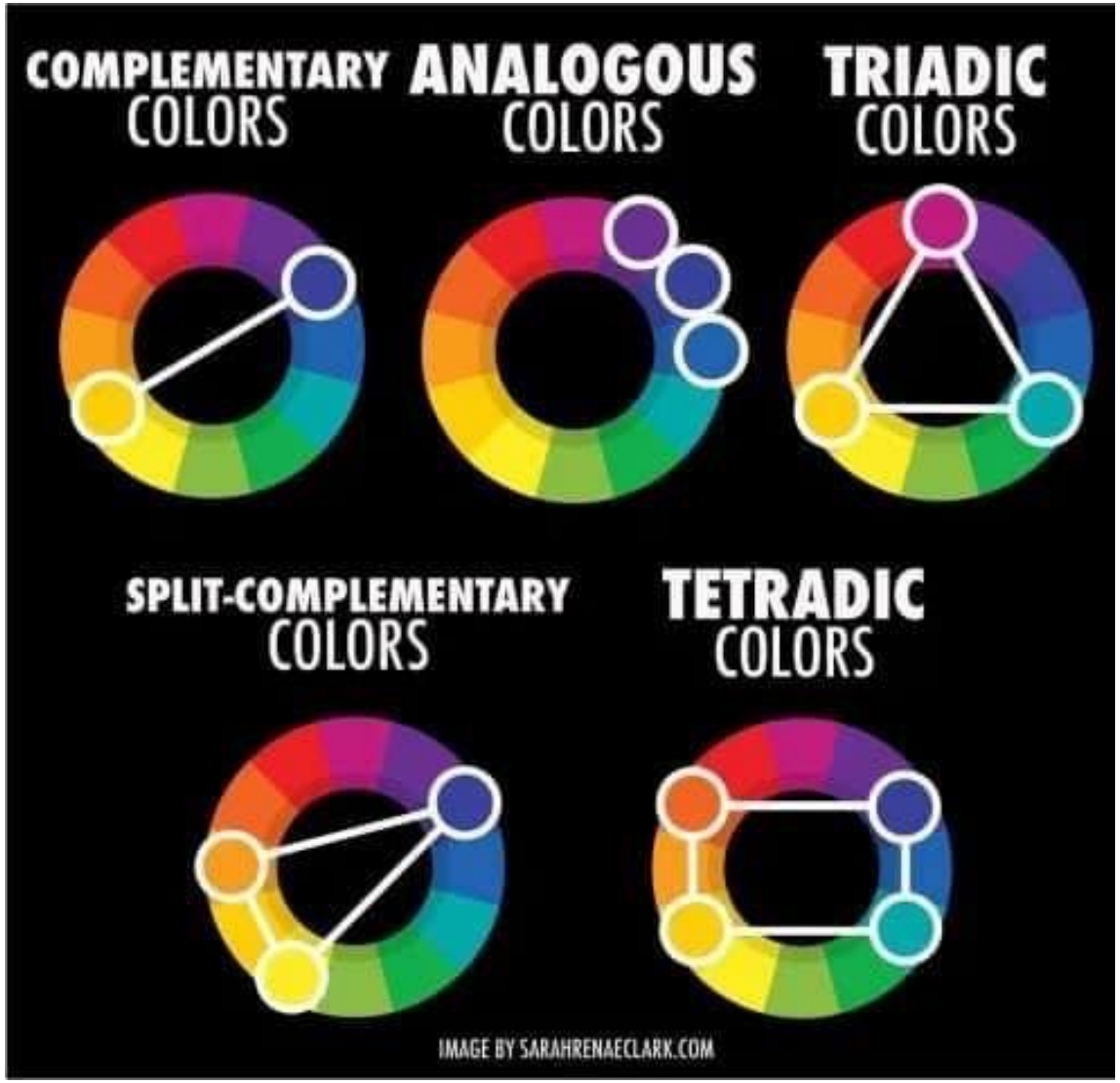
Human-Computer-Interface (HCI) Testing – Tools for Good Practices for Appearance on software/UI

- https://material.io/design
- https://fonts.google.com/icons?icon.style=Filled



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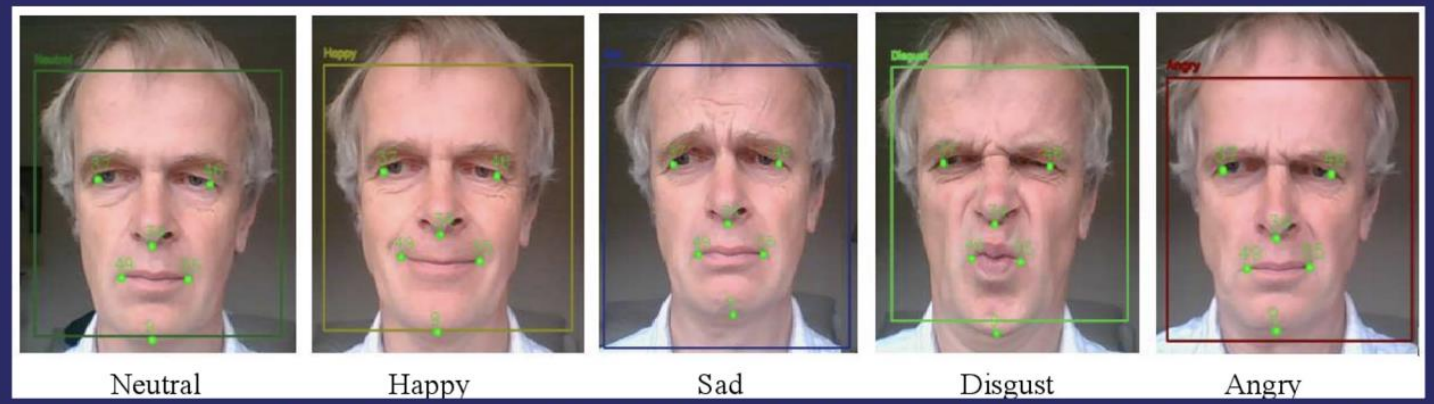
Human-Computer - Interface (HCI) Testing – Device detection on human emotion and gestures

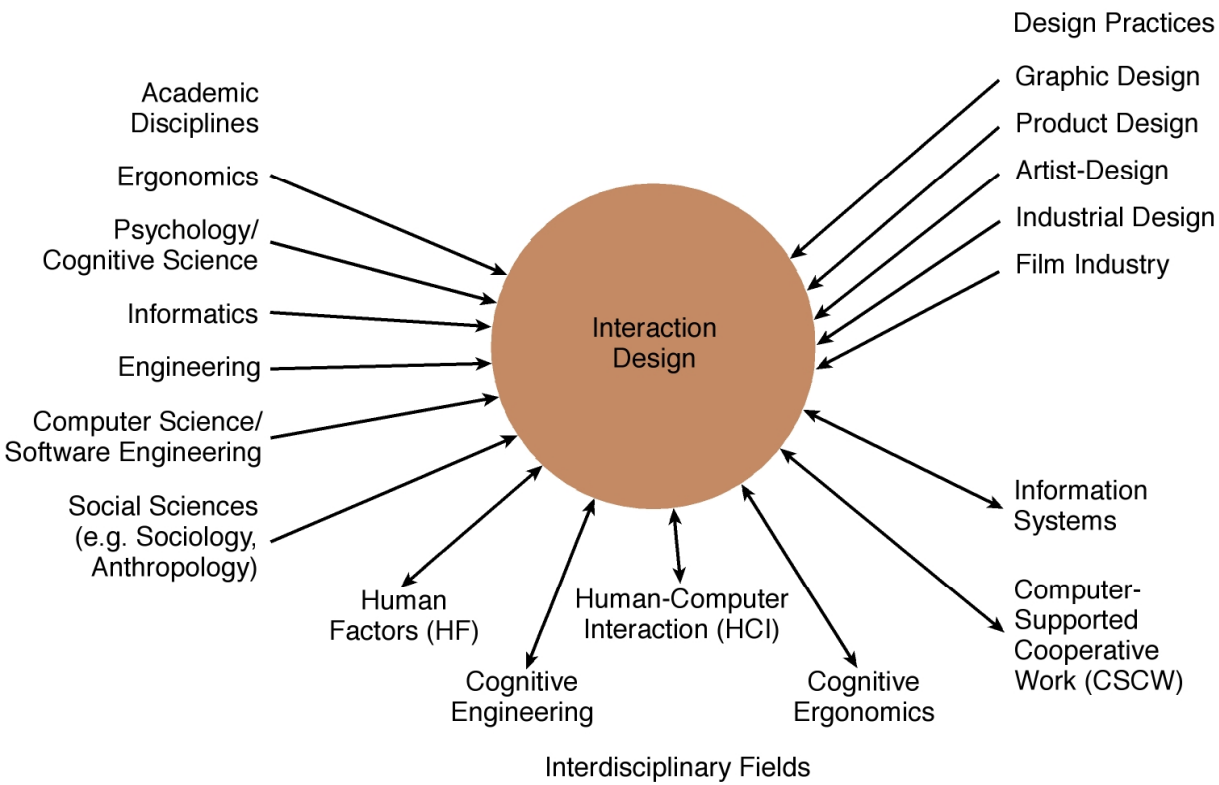
Engagement can be determined from position of pupil, i.e., whether alignment is centered or looking away left/right

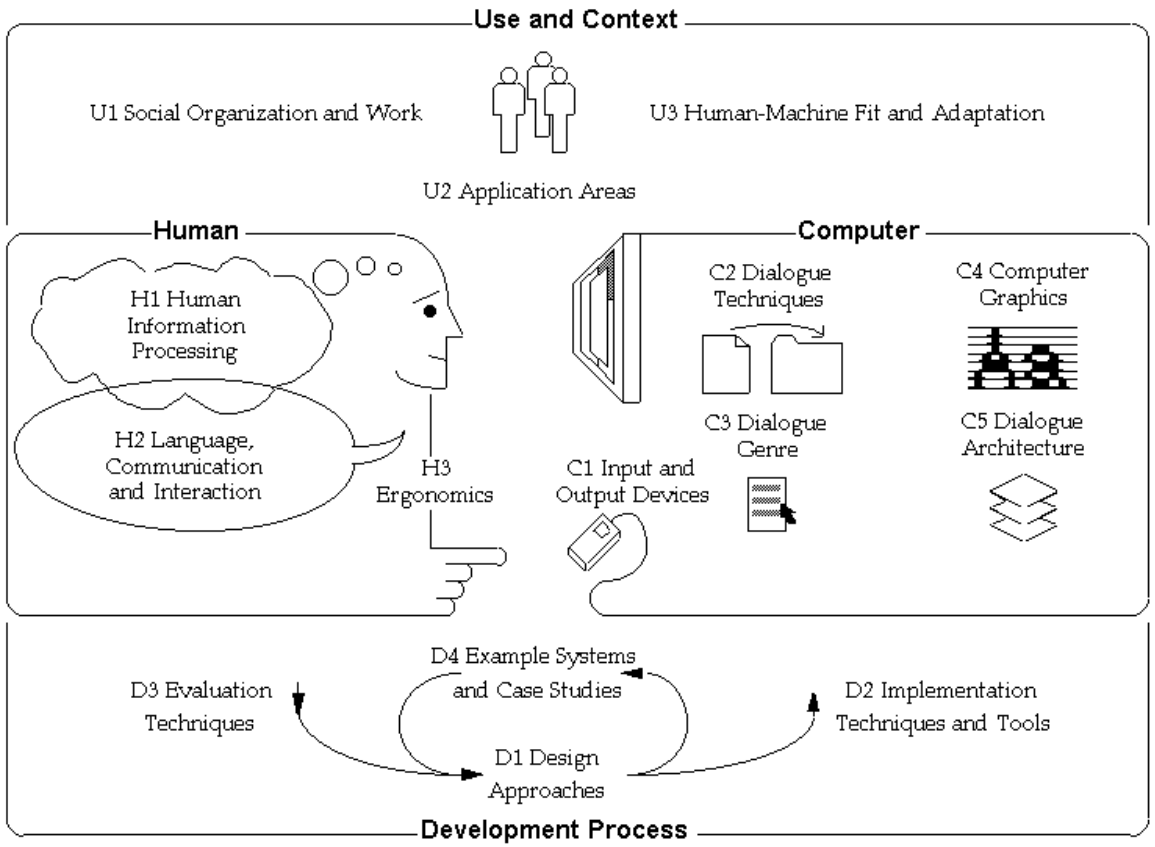


Viewers' moods when viewing content can help advertisers get an even better understanding of engagement with their content

Human-Computer - Interface (HCI) Testing – Usability Goals

Human-Computer -Interface (HCI) Testing – Relationships HCI to other Disciplines



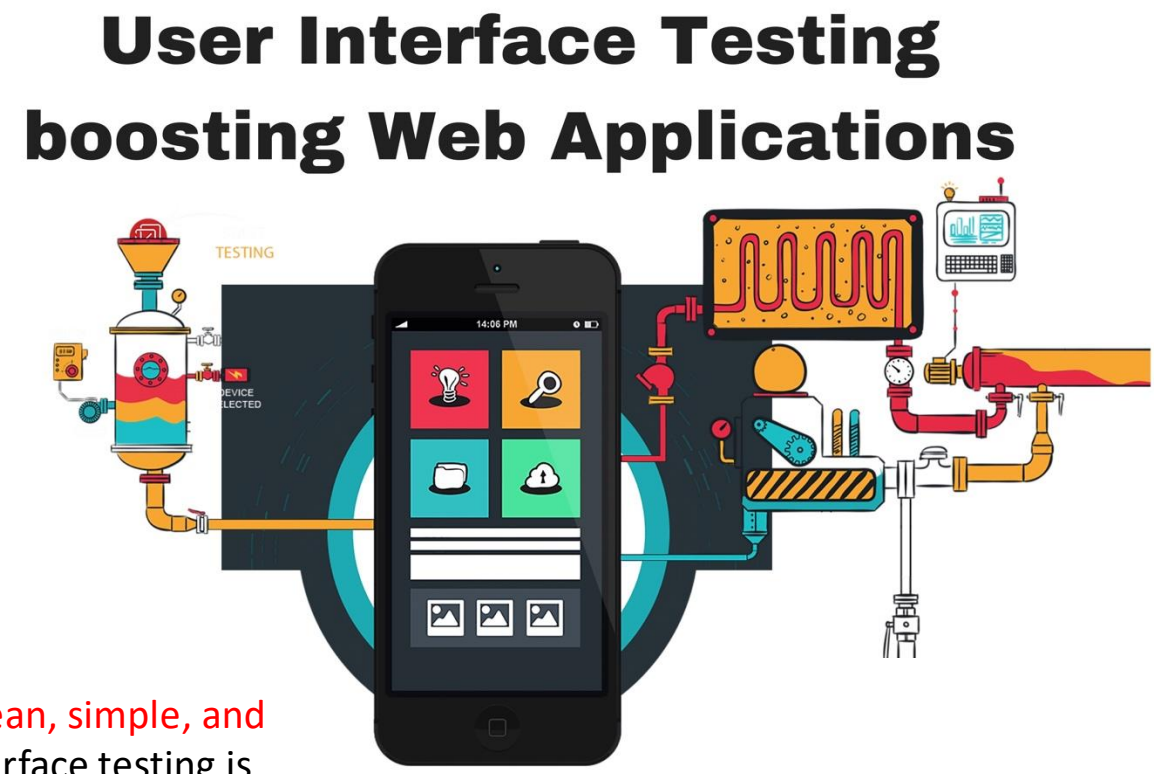


Human-Computer-Interface (HCI) Testing - Testing techniques for web sites

An Overview on the Most Common Bugs Faced in HCI Testing

What is HC Interface Testing?

The HC Interface Testing or Graphical User Interface Testing is performed to check whether all the user interaction functions like the menus, graphic element alignments, images, videos, fonts, colors, forms, etc are working as per the specifications.



Here are the most common HC Interface testing bugs:

Compatibility with devices

Device fragmentation issue is the elephant in the room whenever it comes to developing a website or an application. Device fragmentation is caused due to the incompatibility between the browsers and OS, OS and devices, browsers and OS, screen sizes, form factor, etc.

Responsive design; Layout; Font; Color

HC Interface of a website

-Gives the first impression for users, so we need to make a point that it is clean, simple, and user-friendly. But the HC Interface of a website is most vulnerable as HC Interface testing is mostly done manually hence a lot of human errors can occur.

But with the evolution of HC Interface testing, many advanced automated tools, and test automation frameworks are coming up. So, in order to avoid any errors related to HC Interface testing, one must make use of advanced test automation platforms like pCloudy. pCloudy is a continuous testing platform that provides both manual and automation testing.

Comparison between developed (Australia) & developing (China) countries

The influence of user culture on website usability

Authors:
Rukshan Alexander, Nik Thompson ,
Tanya McGill, & David Murray

Reference:
https://nikthompson.com/PDF/Alexander-Thompson-2021-Culture-and-Website-Usability.pdf

Human-Computer - Interface (HCI) Testing – Current Practices according to countries

Test 2 existing websites:

1. English and translated Chinese versions of an Australian website.

The Department of Finance - Western Australia (2016) website was originally designed in English.

2. Chinese and translated English versions of a Chinese website.

The Beijing Municipal Government Portal (2016) website was originally designed in Simplified Chinese.

(a) Australian version of website design in English (b) Chinese version of website design in Mandarin

Purpose of the article

- To demonstrate the significant interaction of user culture with website usability and satisfaction on a set of translated Australian and Chinese website.

- To investigate how cultural factors and website design interact to affect user performance and satisfaction.

Methodology: Quantitative
Method: Survey
Instrument: Questionnaire

- Participants are asked to complete online questionnaire which obtained demographic data and measured user satisfaction
- 6 items of a composite measure of satisfaction:
  - In general, I am satisfied with the design of the website (1)
  - In general, the browsing experience that I had with the website was satisfactory (2)
  - Using the website made me frustrated (reverse coding) (3)
  -
  - I feel terrible when using the website (reverse coding) (4)
  - After using the design, I will never recommend it to my friends (reverse coding) (5)
  - After using this website, I will never use it again (reverse coding) (6)

Sample size-Respondents

- 100 Australians and 100 Chinese participants.
- All Australian participants are native English speakers living in Australia.
- All Chinese participants are native Mandarin speakers living in China.
- Required to have desktop computers, recruited by emails that contained a survey link with the survey instructions as well as a consent form.

Human-Computer -Interface (HCI) Testing – Current Practices according to countries

- Website usability attributes differ between cultures and culturally specific websites.
- The use of cultural web attributes can increase user perfomance and satisfaction with websites.
- Chinese users show less difference in performance when using non-culturally specific designs than Australian users.
- Internet is a Western creation and though China has a booming tech industry, the major Internet sites are often designed with a Western audience in mind.
- Australian Internet users generally see far fewer Asian designed websites, therefore, the difficulty faced is greater when they are asked to interact with a nonfamiliar culturally specific website.
- A possible future direction (culturally specific designs) may converge and normalize toward a single "tech-culture", which may be closer to a Western design.

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Innovative Method of Data Collection: Think-aloud

Importance of Conducting Think-Aloud in HCI

1

Immediate User Feedback

The think-aloud method in Human-Computer Interaction (HCI) provides immediate insights into users' thought processes, allowing designers to understand user behavior in real time.

2

Identifying Usability Issues

By vocalizing their thought process, users reveal usability issues that may not be apparent through traditional testing methods, enabling designers to identify and address potential problems early in the development process.

3

Enhancing User-Centered Design

Conducting think-aloud sessions fosters a user-centered design approach by placing the user's perspective at the forefront, leading to the creation of more intuitive and user-friendly interfaces.

Steps to Conduct the Think-Aloud Method

1. **Preparation**

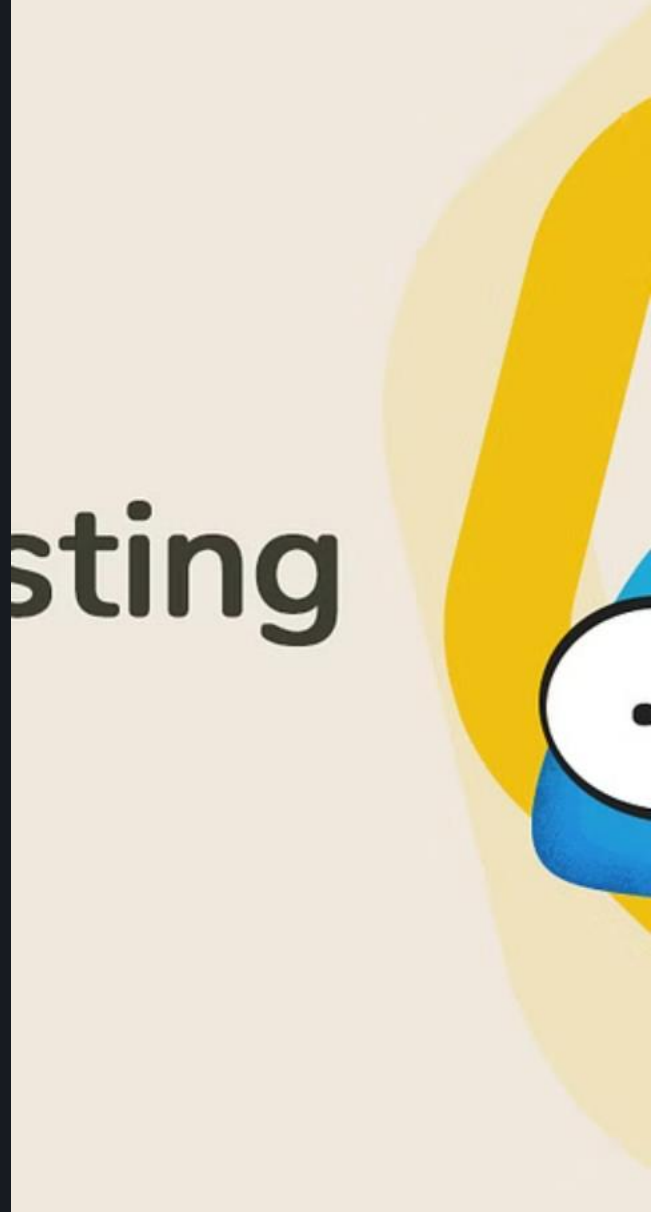
   Before conducting the think-aloud method, define the specific tasks or scenarios that participants will engage with in the usability study. Create a comfortable and distraction-free environment for the participants.

2. **Instruction**

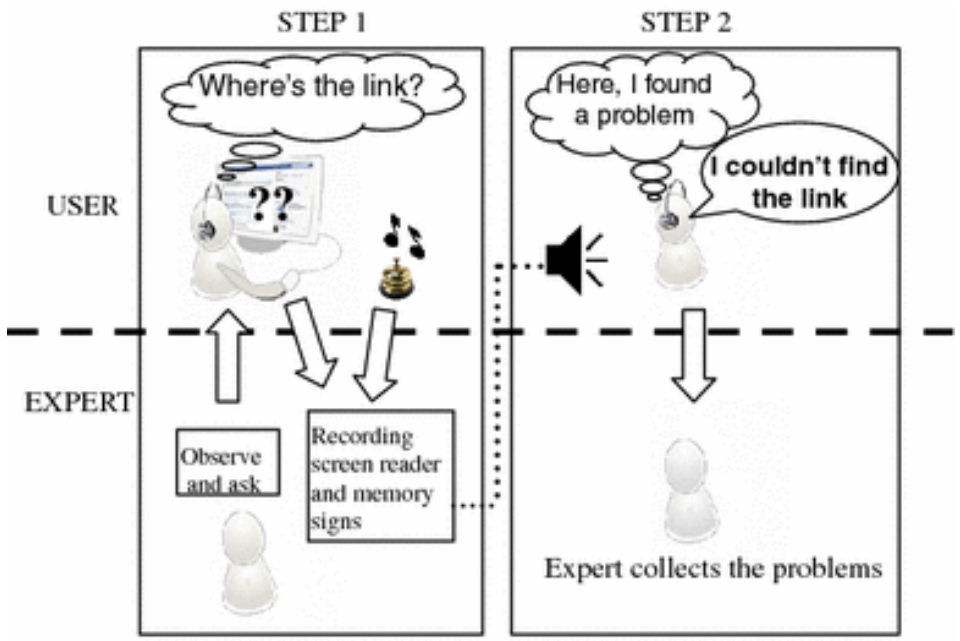
   Provide clear and concise instructions to the participant, explaining the purpose of the think-aloud method and how they should verbalize their thoughts while completing the tasks.

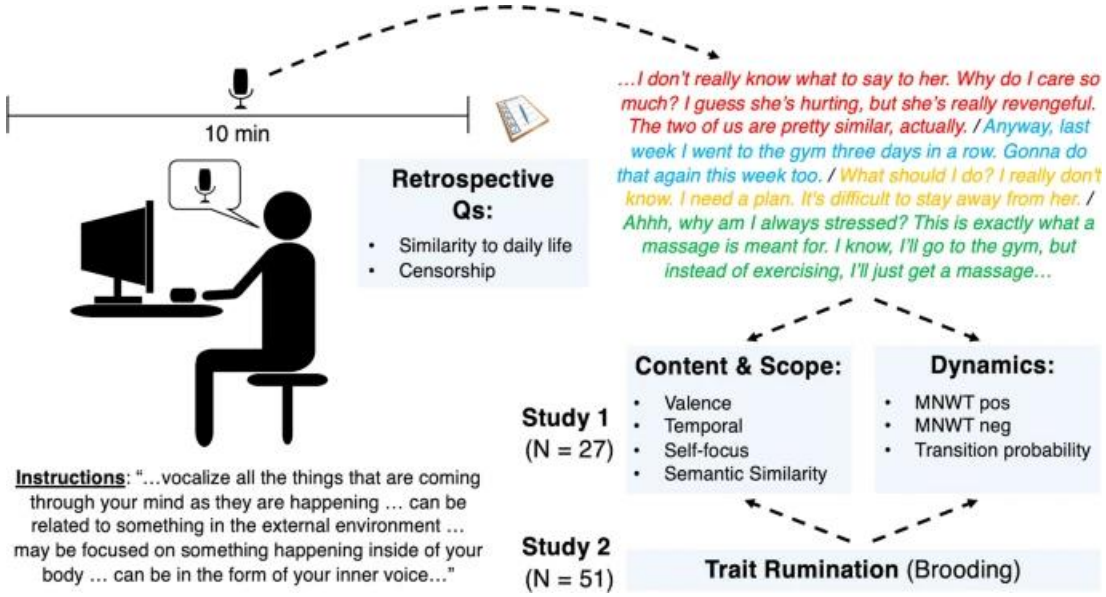
3. **Task Execution**

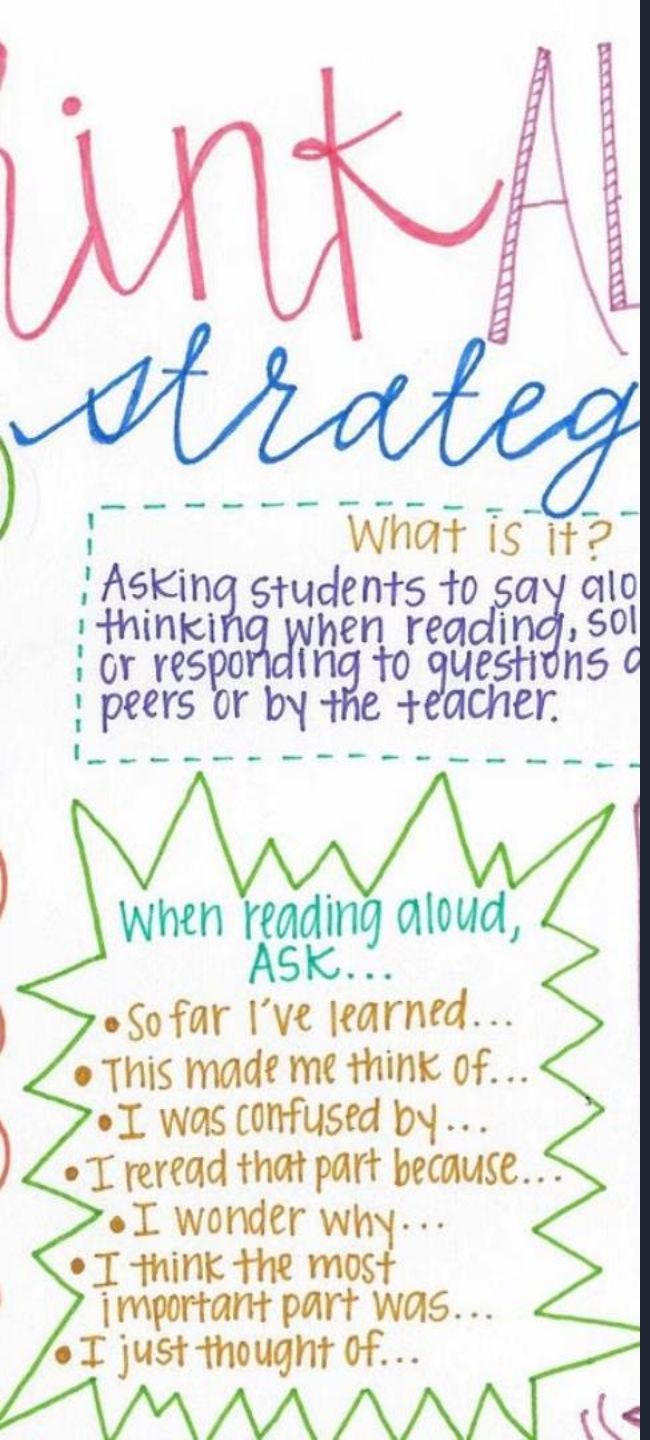
   Observe and document the participants' thought processes as they perform the designated tasks, noting any points of hesitation, confusion, or insights shared during the process.



Human-Computer -Interface (HCI) Testing – Think-aloud Method







Best Practices for Facilitating the Think-Aloud Process

Encourage Verbalization

Encourage participants to articulate their thoughts openly without filtering. Assure them that there are no right or wrong answers, and their feedback is crucial for improving the product.

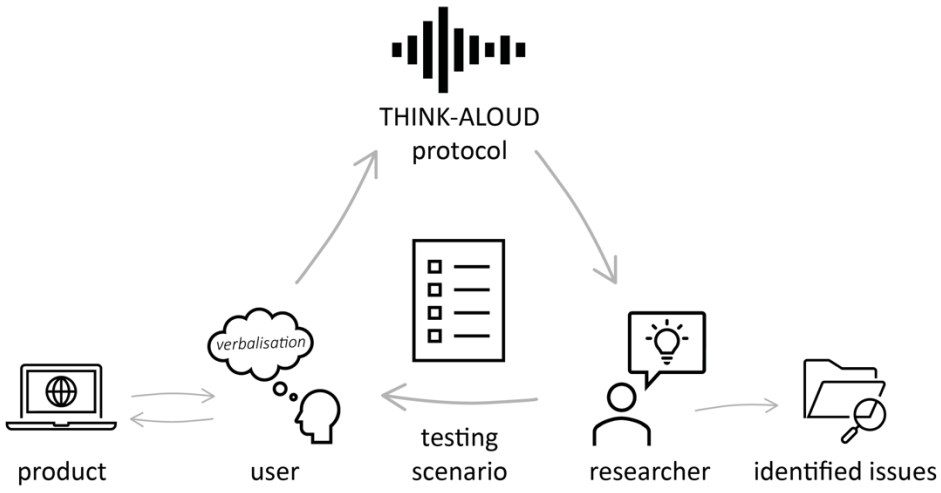
Active Listening

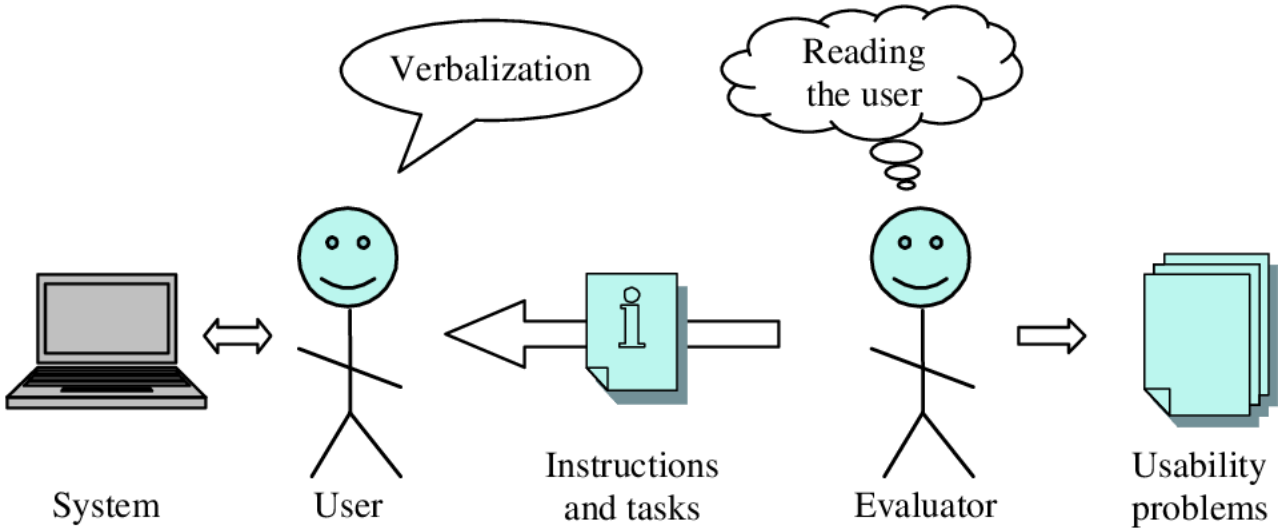
Provide undivided attention to the participants while they verbalize their thoughts. Avoid interruptions and allow them to express their opinions freely.

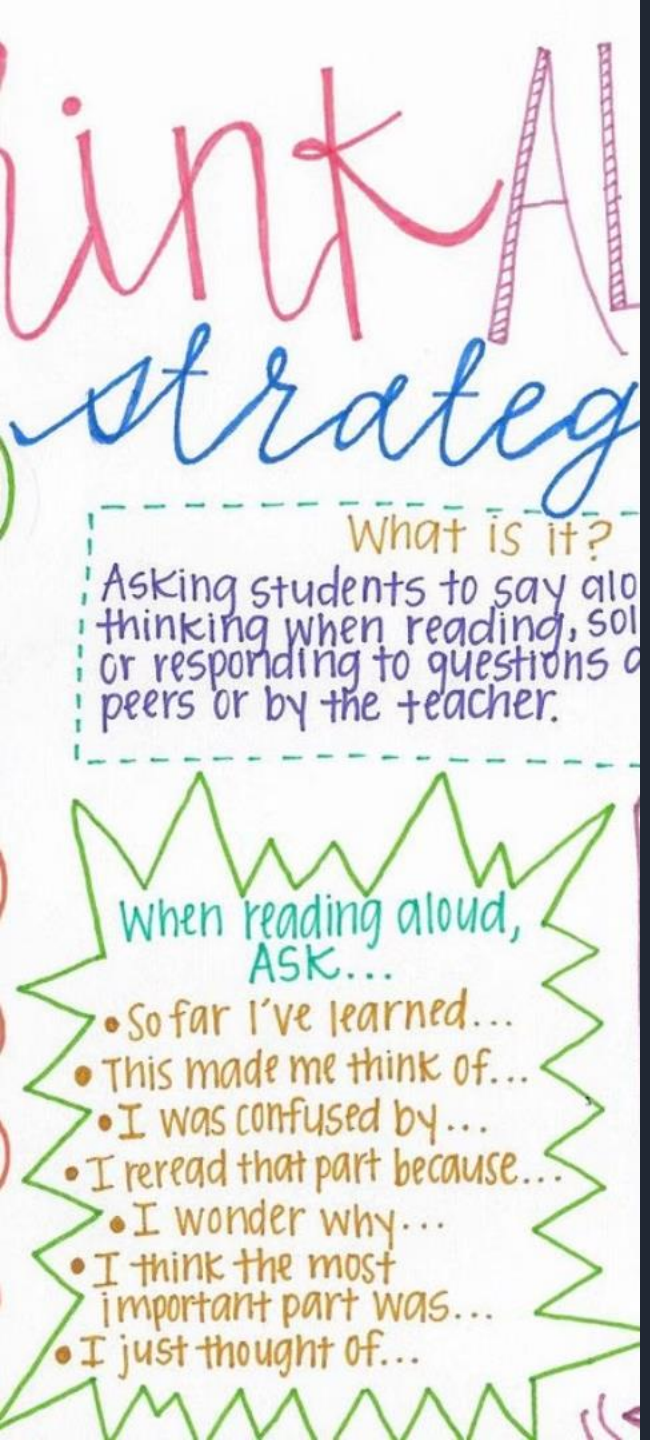
Establishing Rapport

Build a comfortable rapport with the participants to create an environment where they feel at ease and open to sharing their genuine experiences and feedback.

Human-Computer -Interface (HCI) Testing – Think-aloud Method







Common Challenges and How to Overcome Them

1

Overwhelming Silence

If participants experience prolonged periods of silence, gently prompt them to continue verbalizing their thoughts by asking open-ended questions or providing subtle encouragement.

2

Filtering of Thoughts

Participants may initially filter or refrain from sharing certain thoughts. Reassure them that all feedback is valuable and encourage them to express their thoughts freely.

3

External Distractions

Minimize external distractions during the think-aloud session by ensuring a quiet and focused environment. Communicate the importance of concentration to the participants prior to the session.

Analyzing and Interpreting Think-Aloud Data

Data Collection

Thoroughly document the verbalized thought processes and observations during the think-aloud sessions, incorporating both qualitative and quantitative data to gain a comprehensive understanding of the user experience.

Pattern Identification

Analyze the collected data to identify recurring patterns, notable insights, and areas of confusion. This process helps in uncovering common usability issues and user preferences.

Feedback Integration

Integrate the insights gained from the think-aloud sessions into the iterative design process, ensuring that user feedback directly contributes to the refinement of the HCI product or application.

Human-Computer -Interface (HCI) Testing – Example 1

Think-aloud Protocol for Usability Testing (Activity 1-10)

A4: Tasks Sets for: Study 1, An Investigation for the Impact of Task-types on the Reactivity of The Concurrent Think-aloud in Usability Testing

ACTIVITY 1

Home > Holiday > Weddings and Honeymoons
You are planning your forthcoming wedding abroad and have decided to use Thomas Cook. Please find out the phone number of their weddings team.

Please write down the phone number you have found below

.....

ACTIVITY 2

Home > Store locator > Store locator page
You need to ring a local Thomas Cook travel store to enquire for certain information and visit the store for further enquiries. Please find the postcode of Thomas Cook store in address that is in North Shields

Please write down the postcode below

.....

ACTIVITY 3

Home > Luxury > Sri-Lanka
Home > Luxury > Cambodia
Home > Luxury > Thailand

You have decided to go on a holiday to visit either Cambodia, Sri-Lanka or Thailand.

Your decision to choose where to go depends on visa-free entry for British passport holder. Based on the information on Thomas Cook website which country would you visit.

Please write down the country you would visit that has visa-free entry for British passport holder

.....

ACTIVITY 4

Home > Footer > Download brochure > Brochure Store

Your best friend has decided to visit Cyprus, she asked you to download an electronic brochure to help her find out about some of the resorts.

Please find a downloadable link to the electronic brochure that will assist your friend to find out more about Cyprus. Do not use search menu

Please write the name of the brochure below

.....

ACTIVITY 5

Home > Credit card fee > Pricing terms and conditions
You want to book a holiday and pay for it with your Barclaycard. Find out if Thomas Cook make a charge for this

Credit card charge.....

ACTIVITY 6

Home > Flights > Available dates page > Basket > Passengers Information
Please **DO NOT** use the search on the home page, use the navigation menu.

You wish to book a flight from London, Heathrow to Brussels, Belgium on 3rd October 2017 to return 17th October 2017.
You wish to use British Airways

Please first complete the above task then include the following extras on your booking

Extras

i. You would like to add a hotel accommodation offered by Thomas Cook that is close to Brussels City Centre
ii. You would like to add airport transfers for £200 or less
iii. Your total basket price should not be more than £2000

Stop where you are required to enter passenger details

Please tick appropriately where you met the criteria

You use British Airways

Hotel accommodation close to city centre

Airport transfers for £200 or less

Total basket price £2000 or less

ACTIVITY 7

Home > City Escapes > Filter page > Basket > Passengers Information
To mark your birthday, you and your partner have decided to visit Amsterdam in November 2017 for a duration of 5 nights.

i. You wish to depart from London Heathrow using KLM Royal Dutch Airlines
ii. You want a twin standard room that includes bed and breakfast
iii. Upon arrival at Amsterdam, you require Airport transfer on a Luxury Eco Car to your hotel.

Your set budget for this trip is **£990** or less for both of you.
How can you accomplish this task?

Stop where you are required to enter passengers' information.

Please tick the criteria you have met

Depart from London Heathrow using KLM Royal Dutch Airlines

Twin standard room with bed and breakfast

A Luxury Eco Car to your hotel

Set budget for this trip is £990 or less

ACTIVITY 8

Home > Flight > Filter page > Basket > Passengers Information

You are planning a 7nights summer holiday (May 2017) to visit Spain with your partner and two kids 5 and 8 years old that includes flight and hotel payment in one package.

- You wish to depart from London, Gatwick airport
- You want hotel accommodation with safety deposit box

Please first complete the above tasks then include the following extras on your booking

Extras

i. You need a flexible booking in case you would want to change your holiday.
ii. Outbound luggage allowance of 20kg for two adults

How do you accomplish this holiday booking with a set budget of **£2,300** or less?

Please stop where you are required to enter passengers' information.

Please tick appropriately if you were able to meet the following criteria

Flight from London, Gatwick airport

Hotel with safety deposit box

Flexible booking

Luggage allowance of 20kg for two adults

A set budget of £2,300 or less?

ACTIVITY 9

Home > Cruise > Mediterranean > Cruise search results > Cruise finder tool > Cruise search results > Passenger Information

You wish to go on a Cruise to the Mediterranean on August 2017 for a duration of 10 to 14 nights and have chosen Southampton, England to be the port where the cruise ship will depart.

You **DO NOT** need a specific cruise line or cruise ship

i. You want to include airfare to and from Newcastle Intl. Airport, Newcastle
ii. You want a cruise that will visit at least 5 different ports
iii. You want a cabin with balcony and ocean view
iv. The total budget for the booking should be **£4,450**.

How can you accomplish these tasks?

Stop where your booking meets all criteria including your set budget.

Please tick appropriately where you met the criteria

Airfare

A cruise that will visit at least 5 ports

Cabin with balcony and ocean view

A set budget of £4,450 or less

ACTIVITY 10

Home > Extras > Contact Extras

You have recently come back from a cruise holiday. During your time away you became unwell and ended up missing a shore excursion because you were stuck in your cabin for 3 days.

Luckily you took out Thomas Cook Silver insurance

i. See if you can claim the cost of this excursions which was £400
ii. Find out if you can claim any money for the 3 days of your holiday that you missed
iii. Find out the contact number/email for making a claim

Please write you're your answers

Can you claim the £400 cost of your excursions:.....

Can you claim any money for the 3 days holiday you have missed?

.....

Contact number to make a claim

.....

Email address to make a claim

.....

Human-Computer -Interface (HCI) Testing – Example 2

Think-aloud Protocol for Usability Testing (Activity 1-10)

Instruction: I would like you to think-aloud, tell me the things that you like, dislike or find confusing about the site as you complete each task

Activity 1

Find and download the bus timetable for the Arriva bus X18 from Newcastle, Haymarket to Morpeth

Answer Book

Where you able to find and download the time table for Arrival bus X18?

- ☐ Yes
- ☐ No

Activity 2

Find out the cost of a student metro season ticket for one month, which covers all zones.

Answer Book

Please write the cost

.....

Activity 3

Find out if you can park your bicycle at Monument Metro station and the numbers of bicycle racks

Answer Book

Can you park your bicycle?

- ☐ Yes
- ☐ No

Please write down the number of bicycle racks

.....

Activity 4

When does the last ferry leaves North Shields on a Sunday?

Answer Book

Please write the time

.....

Activity 5

You want to apply for a pop card. Rather than apply online or download the application form you prefer to call a team member for further enquiry. Find the phone number of their pop card team

Answer Book

Please write down the phone number

.....

Activity 6

You would like to use the **live travel map** to see the next available Go North East bus 700 from Sunderland University Travel Hub and download the timetable.

How would you accomplish this task?

Answer Book

Please write the time for the next available bus 700

.....

Were you able to download the timetable?

- ☐ Yes
- ☐ No

Activity 7

As a student at the University of Sunderland you usually use the metro to travel from **Newcastle** to **Sunderland**. You would like to get metro alerts via email if any disruptions occur between 7:00 and 15:00 for Mondays and Thursdays. How would you accomplish this task?

Answer Book

Were you able to create an email alert?

- ☐ Yes
- ☐ No

Activity 8

You wish to explore Tyne and Wear to visit places of attraction and have chosen to visit the "Angel of the North" with postcode **NE9 7UB**. Plan your journey from **Sunderland University Metro Station**.

What is the Go North East bus service number with the most convenient route to see the Angle of the North?

Answer Book

Please write down the Go North East bus number

.....

Activity 9

Time and date for this activity is optional

You and your step sister who is 15 years old both live at Jarrow close to the metro station. She will be attending South Tyneside College, South Shields. She has asked you to help her find out:

i. The most convenient route for her journey from home to college
ii. The cheapest return ticket

Answer Book

Write down the most convenient route?

.....

.....

.....

Please write down the name of the cheapest return ticket

.....

Activity 10

You live in **South Hylton** and wants to visit a course mate who lives at **Palmersville**. Since you are going on the metro, you want to buy a return ticket.

i. What are the names of the zones?
ii. What is the ticket's price?

Answer Book

Can you? Please write your answer

Please write down the zones

.....

Please write down the ticket's price

.....

Conclusion and Key Takeaways

1 Continuous Iteration

The think-aloud method serves as a valuable tool for continuous iteration, ensuring that user feedback remains integral throughout the design and development lifecycle.

2 Empathetic Understanding

By fostering empathetic understanding of user experiences, the think-aloud method contributes to the creation of empathetic and user-centric HCI solutions.

3 Enhancing Usability

Implementing think-aloud sessions facilitates the enhancement of usability and user satisfaction, leading to the creation of more intuitive and accessible HCI interfaces.

Made with Gamma

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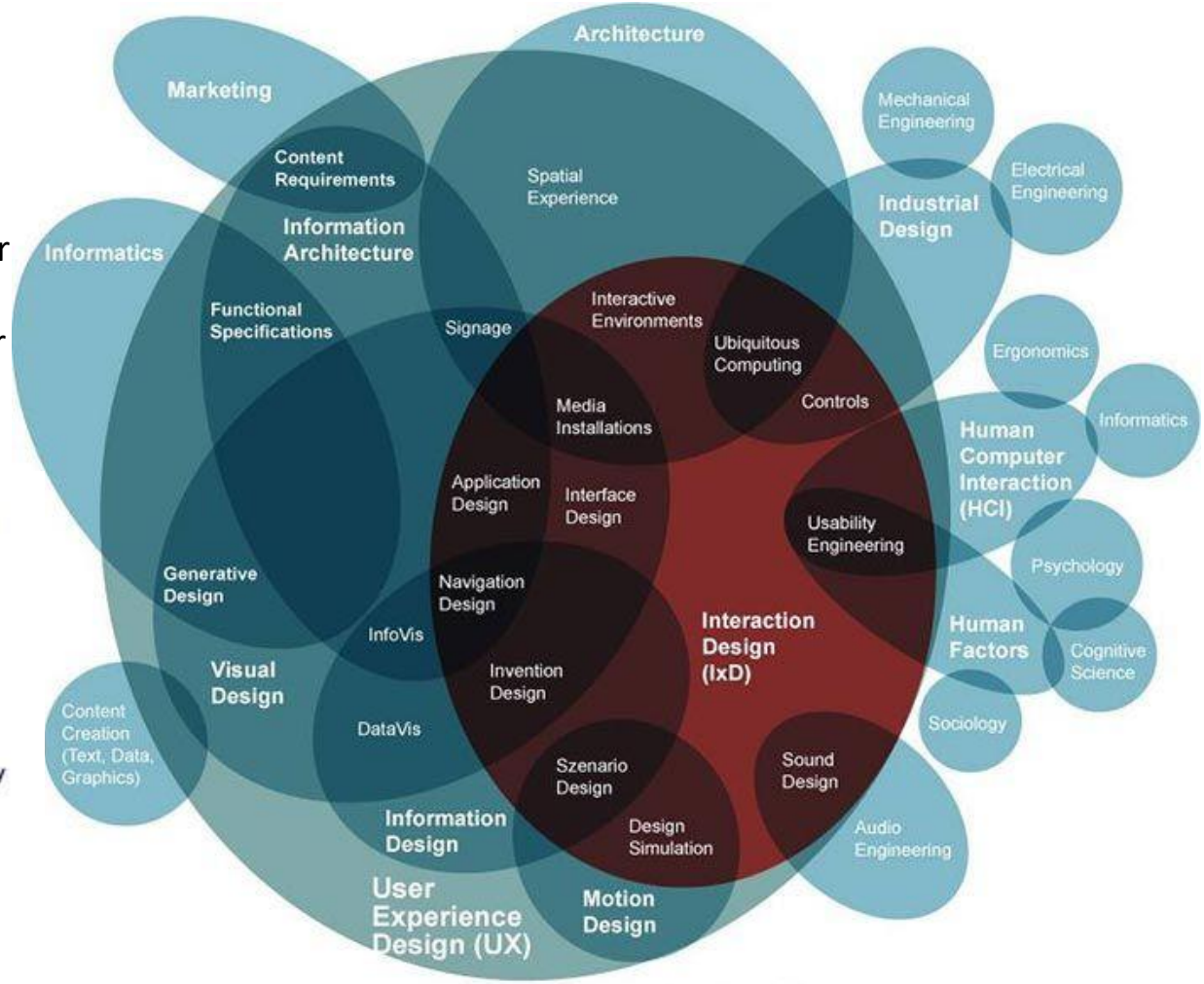
Additional notes…

Human-Computer -Interface (HCI) Testing - The variety of aspects of usefulness and usability, heuristic evaluation, cognitive walkthroughs, user testing approaches (observation sessions, etc.)

Where HCI came from

Until the late 1970s, the only humans who interacted with computers were information technology professionals and dedicated hobbyists. This changed disruptively with the emergence of personal computing in the later 1970s. Personal computing, including both personal software (productivity applications, such as text editors and spreadsheets, and interactive computer games) and personal computer platforms (operating systems, programming languages, and hardware), made everyone in the world a potential computer user, and vividly highlighted the deficiencies of computers with respect to usability for those who wanted to use computers as tools.

Bundesarchiv, Bild 183-1989-0130-010
Foto: Grubitzsch (geb. Raphael), Waltraud | 30. Januar 1989



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based on »The Disciplines of User Experience« by Dan Saffer (2008)
www.kickerstudio.com/blog/2008/12/the-disciplines-of-user-experience

Human-Computer -Interface (HCI) Testing - The variety of aspects of usefulness and usability, heuristic evaluation, cognitive walkthroughs, user testing approaches (observation sessions, etc.)

Purpose

The purpose of this testing method is to validate that the HCI is functioning properly from both the ergonomic and the functional point of view. HCI testing should consider both the input as well as the output boundaries between people, operators and users of systems and the system itself.

Rationale The discipline of HCI deals with the boundary area between humans and engineered systems. More specifically, HCI deals with input devices, which are the means by which humans control systems, and output devices, which are the means by which humans interpret systems information.
HCI performance determines how easily a user may control and comprehend underlying functions of a given system. HCI often is the part of the system that determines the acceptability of the system by end users.

Method Humans control systems by issuing appropriate commands. The outputs of these systems, stemming from these commands, are then monitored. Commands may take many forms, including thrown switches, keyboard strokes, mouse moves, screen touches and voice commands.

When one considers the testing task of HCI he or she should be aware of the range of devices, 50 used for interfacing with systems (see, e.g., Table 5.3). Test engineers must take into account that each and every interface device connected to the system may introduce a certain problem, distinctive to the given device. For example, switches and buttons may typically introduce timing errors or wrong sequence phenomena, keyboards may introduce text string errors, or a display may show incorrect information.

TABLE 5.3 Range of Selected HSI Devices

Input Devices	Output Devices	Combined Input/ Output Devices
• Switches or buttons • Electronic pen or tablet • Joysticks • Mouse • Keyboards • Microphones	• LEDs • Displays or Cathode Ray Tubes (CRTs) • LCDs • Head-Up Display (HUD) • 3D goggles • Earphones or speakers	• Touch displays

As can be seen many, HSIs are uniquely designed to meet the needs of specific applications. For example, Figure 5.10 depicts a ruggedized package of switches, lights, keyboard and display typically used in aircrafts, mobile control centers, and Computer Numerical Controlled (CNC) machine tools.

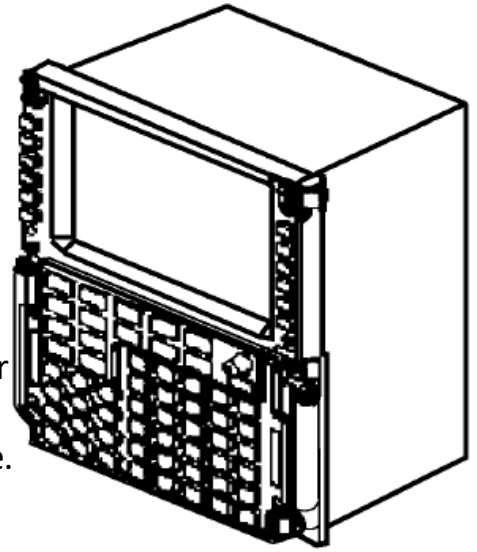

Figure 5.10 Example of an HSI device.

50 Of course, there is a broad range of other engineered systems with their own specialized Input/ Output (I/O). A shower stall is an engineered system with human input consisting of countless types of faucets and we use our sense of sight and touch in lieu of a system output device.

VERIFICATION, VALIDATION, AND TESTING OF ENGINEERED SYSTEMS - AVNER ENGEL Pg 373-378.

Human-Computer-Interface (HCI) Testing - The variety of aspects of usefulness and usability, heuristic evaluation, cognitive walkthroughs, user testing approaches (observation sessions, etc.), Testing techniques for web sites and formal experiments to test hypotheses about specific HCI controls.

HSIs for Automatic Systems

We used these general principles along with the technical basis derived from the literature to develop design review guidelines. The guidelines for reviewing the HSIs to automation are divided into the seven (7) sections shown in the Table A, brief description of the topics addressed is also provided in the Table A. *(pg 7-11)*

Overall Representation of an Automation System

The HSI should accurately represent automation and its plant interfaces. Additional Information: Providing a representation of the automation and the aspects of the plant with which it interfaces helps operators to link the actions of automation to its goals for the plant itself. For example, if automation is maintaining a level in a tank that has a leak, so long as automation can pump water in, the level is achieved, and operators may not know there is a problem. When the level can no longer be maintained, operators need to quickly determine whether the failure is in the automation or the controlled system. Offering an overall representation of both automation and its plant interfaces helps operators assess this situation.

Table A: Guidance for HSIs for Automatic Systems

Guidance Section	Topics Covered
1 Automation Displays	
1.1 General Display Considerations	This section addresses the use of hierarchal access to information so operators can see automation's status at a quick glance and access more detailed information as necessary.
1.2 Automation Representation	This section addresses the overall representation of an automation system, including its current goals, processes, inputs used, current responsibilities of all human and automatic agents, and automation's reliability.
1.3 Automation's Dynamic Status	This section addresses the display of automation's progress on goal and task accomplishment and support of the prediction of future performance. It also addresses notifications such as to the need for automation, significant status changes, and failure of automation initiation.
2 Interaction and Control	This section addresses the automation-operator interaction, including operator control of interaction pace and the need for functionality to support all operator interactions with automation, such as feedback for operator inputs and support for operator queries. The section addresses automation's communications with operators as well.
3 Automation Modes	This section addresses the need to indicate automation's mode and its implications for human role changes. Also addressed are controls for mode changes, notification of automatic mode change along with the conditions leading to automatic mode change.
4 Automation Levels	
4.1 Shared Control	This section addresses the information needs for coordinating tasks which operators and automation share.
4.2 Operation by Consent	This section addresses providing information on task status, the information needed to authorize task continuation, and identifying valid options.
4.3 Operation by Exception	This section addresses providing information on the reasons for the exception and identifying valid options.
5 Adaptive Automation	This section addresses the use of predefined sets of roles and responsibilities to manage adaptive automation. It also provides guidance on the control of automation shifts by operators and by conditions that trigger automatic shifts. Notification of impending shifts and shift conformation is also addressed.
6 Error Tolerance and Failure Management	This section addresses the alerts to automation failures and degraded conditions. In addition, guidance is provided for providing information on the causes of failures and for the support of failure recovery. The monitoring of personnel actions is also addressed.
7 HSI Integration	This section addresses the integration of automation HSI's with existing HSIs, procedures, and supporting reference materials.

Human-system Interfaces for Automatic Systems - John O'Hara and James Higgins Brookhaven National Laboratory Stephen Fleger and Valerie Barnes, U.S. Nuclear Regulatory Commission Pg 7-11.

Human-Computer -Interface (HCI) Testing - The variety of aspects of usefulness and usability, heuristic evaluation, cognitive walkthroughs, user testing approaches (observation sessions, etc.)

Processes of Automation

Automation uses input from the plant (and perhaps the operator) and processes the information to accomplish a goal. These processes are an important aspect of automation in that they are the means by which automation performs its tasks.

Modes of Automation

Automated systems may have different modes of operation. Modes define sets of mutually exclusive behaviors that describe the relationship between input to the automation and the response to it (Jamieson & Vicente, 2005).

Flexibility of Allocation

A system can be designed such that the human or machine agent responsible for performing an activity is always the same, i.e., static allocation. Alternatively, a task can be performed either by automatic systems or by personnel based on situational considerations, such as the operator's overall workload.

Reliability of Automation

The final dimension of automation is reliability. All engineered systems have less than perfect reliability. Automatic systems can fail in whole or in part and thus compromise their ability to achieve their intended function. When an automatic system has a simple, well-defined task to accomplish, its reliability is easy to quantify, e.g., as the probability the system will correctly perform its function.

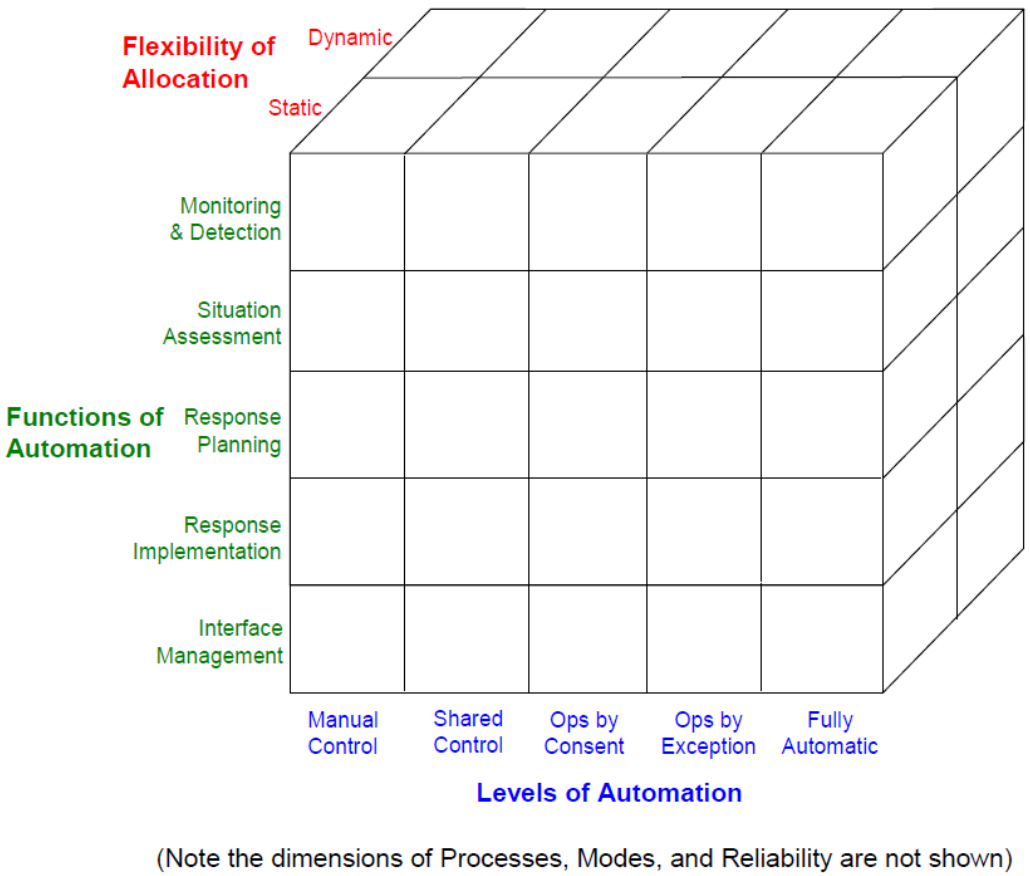

Figure 2. Framework for characterizing automation

Conclusions: This characterization of automation is based on six dimensions. Figure 2 illustrates three of the six dimensions, along which all automatic systems can be characterized.

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Thank You…
Terima kasih..